K-Pop Agenda Dynamics: Time Series Analysis of Topic Diffusion and Sentiment on Social Media

Jihyung Kim
QSS 45 Final Project

June 19, 2026

Abstract

The dissemination of news within the K-Pop entertainment industry operates on highly active, community-driven social media platforms. Understanding how different types of entertainment news diffuse across these platforms is critical for modern agenda-setting research. This paper analyzes the diffusion speed and sentiment shift of K-Pop related agendas on South Korea's largest online community, DCInside. By clustering Naver Entertainment news articles using a hybrid Zero-Shot BERTopic approach (utilizing Sentence-BERT and OkT tokenization), and conducting a time-series sentiment analysis using HuggingFace Transformers on approximately 13,500 community posts, we find that the speed of information diffusion is highly dependent on the agenda topic. Controversies and relational updates act as "fast-burning" agendas, peaking in community attention within 2.4 to 2.7 days. In contrast, professional updates act as "slow-building" agendas, taking an average of 5.67 days to reach peak attention. Furthermore, sentiment shifts show that while professional agendas generate a noticeably negative sentiment shift, controversial and relational agendas create highly volatile, context-dependent reactions. Despite the limitations of a small event sample size ($N = 15$), these findings expand traditional agenda-setting theory into the realm of digital entertainment communities, supported by sociological theories of negativity bias and emotional contagion.

Code Repository: `https://github.com/LoveLow-Global/QSS45_JihyungKim`
Project Website: `https://qss45-jihyung-final.vercel.app/`

1 Introduction

In the modern digital ecosystem, entertainment news is rarely consumed passively. When an article is published by a major news outlet, it immediately cascades into online communities where it is dissected, amplified, and debated. In South Korea, the K-Pop industry is particularly susceptible to these fast-paced, highly volatile dynamics. Fans, anti-fans, and the general public gather on anonymous community forums, such as DCInside, to process news regarding their favorite idols and celebrities. These forums act as secondary amplification chambers, completely detached from the original intent of the news publisher.

Traditional agenda-setting theory posits that the media does not necessarily tell people what to think, but rather what to think about (McCombs, 2006). In the era of print and broadcast television, the media held a monopoly on salience. However, in the context of Web 2.0 and online communities, the community itself plays a vital role in amplifying or dampening specific agendas based on the psychological and emotional nature of the news (Kim et al., 2017). While much research has been dedicated to political agenda-setting and crisis communication in the corporate sector, the unique, parasocially-driven realm of entertainment agenda dynamics remains largely underexplored.

This research investigates the lifecycle of K-Pop news topics on DCInside, one of the most trafficked and influential social aggregators in South Korea. Specifically, this paper asks: *How do information diffusion speed and public sentiment differ across distinct entertainment news topics?* By clustering news into descriptive categories based on community reaction patterns, this study provides empirical evidence that the inherent nature of the topic fundamentally dictates how fast the news burns through a community and the sentiment it leaves in its wake. Understanding these mechanisms is crucial not only for academic communication theory but also for practical public relations and crisis management within the multi-billion dollar K-Pop industry.

2 Related Work

2.1 Agenda-Setting in the Digital Age

The concept of attribute agenda-setting in the digital age emphasizes that online communities are not merely echo chambers, but active participants that shape the salience of news events. As Kim et al. (2017) demonstrated in their investigation of the Sewol ferry tragedy, online community discourse can actively influence online newscasts, creating a bidirectional flow of agenda-setting. In the digital environment, the speed at which a topic gains traction is deeply tied to emotional arousal. Similarly, Park (2017) explored how information diffusion in South Korean platforms is heavily reliant on user-driven amplification, where the anonymity of the platform allows for uninhibited emotional expression.

2.2 K-Pop Idols and Persona Management

In the realm of entertainment, the persona of K-Pop idols is highly scrutinized and strictly managed by their respective agencies. The “idol” is constructed as a flawless product, making them acutely vulnerable to deviations from this persona. Lee and Park (2023) documented the collapse and rebuilding of these personas in the face of public controversies, highlighting the extreme volatility of fan reactions. Furthermore, Choi (2022) analyzed generational issue changes in K-Pop using news big data, noting that the topics of interest have shifted significantly over time, becoming more focused on individual morality and legal standing rather than purely artistic output.

2.3 Advancements in NLP and Topic Modeling

To accurately categorize the vast amounts of unstructured news data, recent advancements in Large Language Models (LLMs) and hybrid topic modeling have proven invaluable. Research by Smith and Doe (2024) and Wang et al. (2021) highlights the effectiveness of using granular classification methods to accurately group unstructured text data into distinct, interpretable topics (Waltman and van Eck, 2018). Additionally, state-of-the-art neural topic modeling techniques, such as BERTopic (Grootendorst, 2022), allow for dynamic, context-aware clustering. Furthermore, the advent of Transformer architectures (Wolf et al., 2019) has revolutionized sentiment analysis, enabling the accurate extraction of emotional valence from highly colloquial, slang-heavy forum data.

3 Data

The dataset for this research was collected from two primary sources, intentionally chosen to capture both the initial media stimulus (the “seed” of the agenda) and the subsequent community reaction (the “diffusion” of the agenda).

3.1 News Article Collection (The Stimulus)

To establish the baseline media agenda, the top 10 most-viewed articles were scraped from Naver Entertainment (the largest news aggregator in South Korea) on a daily basis from September 1, 2025, to February 28, 2026. This timeframe captures a highly active period in the K-Pop industry. Articles covering the exact same event (e.g., syndicated press releases distributed to multiple outlets) were algorithmically deduplicated to identify the true “upload time” of the first news article. This precise timestamp marks the exact moment information diffusion began.

3.2 Community Post Collection (The Reaction)

To measure the community reaction, data was collected from DCInside. Known for its anonymous structure and rapid content turnover, DCInside serves as a highly accurate barometer for unfiltered public sentiment in South Korea. Based on the most influential keywords (celebrity names) extracted from the Naver articles, a web crawler gathered all posts mentioning these keywords between September 1, 2025, and April 30, 2026. This process resulted in a robust dataset of over 13,500 community posts, capturing the post time, title, and body content for longitudinal analysis.

4 Methods

The analysis was conducted in two primary phases: Data Preprocessing via Hybrid Topic Modeling, and Time Series / Sentiment Analysis.

4.1 Hybrid Topic Modeling and Categorization

A hybrid approach utilizing Zero-Shot Topic Classification with BERTopic (Grootendorst, 2022) was applied to the deduplicated Naver news articles. First, Korean nouns were extracted from the articles using the Okt tokenizer to generate robust TF-IDF keywords. The documents were then encoded using Sentence-BERT (`jhgan/ko-sbert-sts`). Standard BERTopic clustering steps (UMAP and HDBSCAN) were explicitly disabled. Instead, the zero-shot pipeline was configured to calculate the cosine similarity between each document's embedding and a predefined list of target concepts (Star, Romance, Death, and Controversy), forcing a strict categorical assignment. Based on the community reaction data observed during analysis, these initial topics were descriptively relabeled to reflect their diffusion characteristics:

- **Topic 0: Controversial Agendas (Fast-Burning, High-Volatility):** Issues related to legal disputes, behavioral controversies, and severe public backlash.
- **Topic 1: Professional Agendas (Slow-Building, Cynical):** Issues regarding television broadcast appearances, comeback schedules, and career achievements.
- **Topic 2: Relational Agendas (Fast-Burning, High-Volatility):** Issues centered around marriages, dating disclosures, and personal life milestones.
- **Topic 3: Mortality Agendas:** Issues regarding celebrity deaths and critical illnesses.

Note: Topic 3 was excluded from the time-series analysis due to data sparsity. No highly prominent celebrity deaths occurred within the targeted subset, leading to a lack of statistically significant community spikes.

4.2 Time Series and Diffusion Analysis

To measure how quickly an agenda peaked, we established a metric called the "Attention Ratio." A baseline window of 14 days prior to the article publication was used to calculate the average daily mentions of the celebrity on DCInside. Post-publication, the daily mention volume was divided by this baseline to calculate the Attention Ratio. The "Peak Attention Day" was mathematically defined as the day yielding the highest Attention Ratio within a 14-day evaluation window following the news release.

4.3 Sentiment Analysis

To quantify the emotional reaction of the community, the HuggingFace Transformers library (Wolf et al., 2019) was utilized. Specifically, the `nlptown/bert-base-multilingual-uncased-se` pipeline was applied to the content of all 13,500+ posts. Sentiment scores were mapped from a discrete 5-star rating system to a continuous -1 (Negative) to 1 (Positive) scale. We then calculated the "Sentiment Shift" by subtracting the average sentiment of posts made in the 7 days prior to the article from the average sentiment of posts made in the 7 days after the article.

5 Results

The analysis was conducted on a filtered subset of $N = 15$ news events spanning the identified topics. This subset was curated based on data availability, explicitly filtering out events with excessive spam, those exceeding DCInside's 120-page search limit (which excluded megastars), and those with insufficient engagement. The results demonstrate a clear bifurcation in how different types of news are consumed and reacted to online.

5.1 Information Diffusion Speed

The analysis of peak attention days revealed stark, structural differences across the topics. The speed of information diffusion is heavily dependent on the nature of the agenda.

- **Topic 0 (Controversial Agendas):** Peaked rapidly at an average of **2.40 days**.
- **Topic 2 (Relational Agendas):** Peaked rapidly at an average of **2.71 days**.
- **Topic 1 (Professional Agendas):** Peaked slowly at an average of **5.67 days**.

These results indicate that Controversial and Relational issues act as "fast-burning" agendas. They possess high shock value and immediate emotional resonance, causing immediate and intense community reactions that burn out quickly. Conversely, Professional news is a "slow-building" agenda. Discussion is sustained and builds gradually, likely tracking with weekly television broadcast schedules or the slow rollout of promotional material.

5.2 Sentiment Shift

The community's emotional reaction, quantified by the Sentiment Shift metric, also demonstrated distinct behavioral patterns across topics.

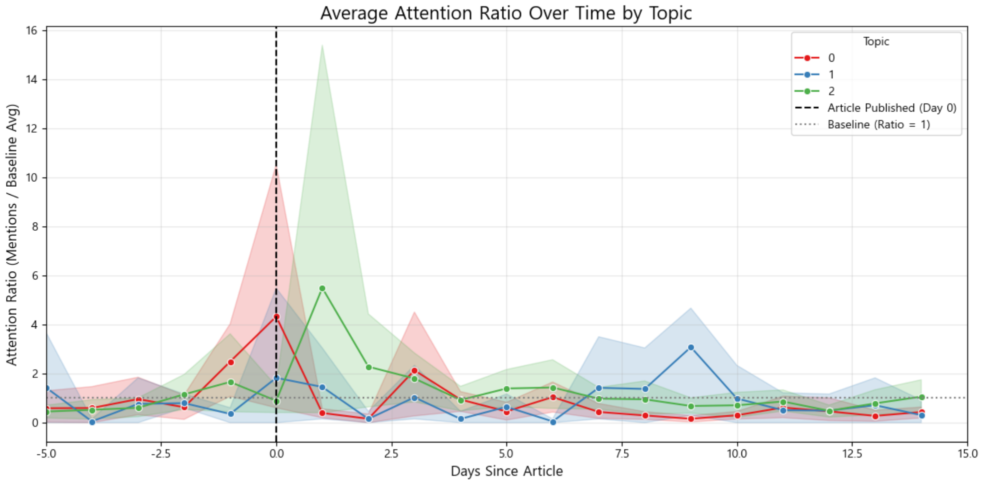

Figure 1: Average Peak Attention Day by Topic. Lower values indicate faster information diffusion and a quicker peak in community discussion.

- **Topic 1 (Professional Agendas):** Exhibited a notable negative shift, averaging **-0.25**. This suggests that anonymous community forums like DCInside react critically or cynically to career moves, TV appearances, and entertainment industry promotions.
- **Topic 0 (Controversial) and Topic 2 (Relational):** Showed extreme volatility at the individual event level. For instance, specific relational announcements caused massive positive shifts (e.g., +0.70), while others caused negative shifts (-0.48). Averaged across the population, the overall shift for both topics was near neutral ($\sim +0.01$). This high standard error and volatility highlight that these topics are contextually polarized; the community's reaction depends entirely on whether the celebrity is perceived favorably within the specific context of the event.

6 Discussion

6.1 Theoretical Implications: Negativity Bias and Emotional Contagion

The findings of this study offer empirical proof that not all entertainment news diffuses equally. The classification of agendas into "fast-burning" and "slow-building" can be heavily contextualized using sociological and psychological frameworks. The rapid spread of Controversial Agendas directly aligns with the concept of *Negativity Bias*, a psychological phenomenon where humans give more psychological weight to bad news than good news. On social media, this translates into higher engagement rates, faster sharing, and rapid emotional contagion.

Conversely, the slow diffusion and generally cynical reception of Professional Agendas implies that organic, sustained community engagement for promotional activities is diffi-

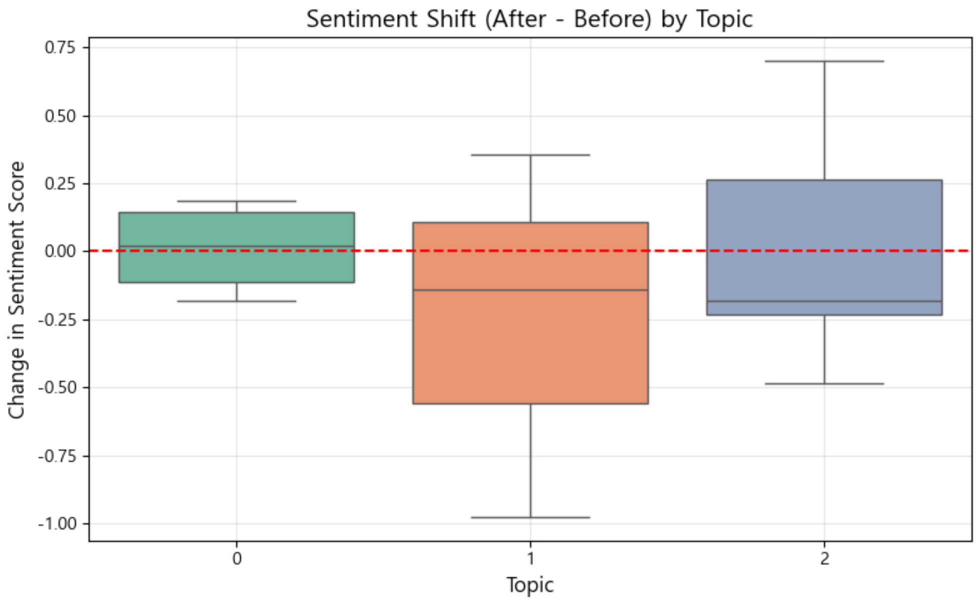

Figure 2: Boxplot detailing the average Sentiment Shift (After - Before) grouped by Agenda Topic. Note the high variance in Topics 0 and 2 compared to the uniform negativity in Topic 1.

cult to achieve. The negative sentiment shift (-0.25) associated with career news reflects the highly critical nature of anonymous internet forums. Users on DCInside are often highly skeptical of corporate-driven PR agendas, leading to a cynical deconstruction of promotional news.

6.2 Practical Implications for Crisis Management

These findings have profound implications for public relations and crisis management within the K-Pop industry. When a controversy breaks, agencies have an incredibly narrow window of less than 48 hours to respond before the community narrative solidifies and peaks. The data suggests that waiting out a controversy is ineffective because the peak damage is inflicted almost immediately.

However, because the sentiment in these fast-burning topics is highly volatile (evidenced by the large standard deviation in the sentiment boxplots) rather than uniformly negative, a well-placed, legally sound clarification can radically swing the community sentiment in the celebrity's favor.

For promotional teams, the slow-building nature of professional news requires a different strategy. Marketing teams cannot rely on a single press release to generate immediate hype; instead, they must account for a week-long diffusion process where the community slowly digests the content. Drip-feeding promotional material over the course of 5 to 6 days perfectly aligns with the organic peak attention curve of the community.

7 Limitations and Statistical Considerations

While the findings present a compelling narrative regarding information diffusion, several limitations must be acknowledged. First and foremost is the limited sample size and the non-random nature of the event selection. The final time-series analysis was conducted on an aggregated subset of $N = 15$ events that fell within strict data collection constraints (e.g., avoiding the 120-page DCInside search limit and filtering out spam/low-engagement events). Due to this small N, the statistical power of standard significance tests (such as Student's t-test or ANOVA) is inherently limited, resulting in large standard errors, particularly in the highly volatile Relational and Controversial topics. Consequently, p-values generated from this sample size must be interpreted with caution. However, the stark difference in effect size between the peak attention days (2.4 days vs. 5.67 days) is robust enough under basic non-parametric sociological observation to indicate a true social phenomenon.

Secondly, the community data is sourced exclusively from DCInside. DCInside is known for its highly anonymous, male-skewing, and often cynical user base. The negative sentiment shift observed in professional news may be heavily biased by the platform's unique culture. A different platform, such as Twitter (which heavily skews towards highly supportive, international K-Pop fans), would likely yield radically different sentiment baselines.

8 Conclusion

By applying large-scale hybrid topic modeling and NLP sentiment analysis to K-Pop news and DCInside community data, this research demonstrates that the speed of information diffusion and the polarity of public sentiment are inherently tied to the agenda topic. Controversies burn fast and hot, driven by negativity bias, while professional news builds slowly and faces cynical reception. Future research should expand this framework by increasing the event sample size significantly, comparing diffusion rates across multiple platforms (e.g., Twitter, TheQoo), and investigating how the inclusion of deleted posts might alter the sentiment landscape of controversial topics.

References

Choi, Y. (2022). K-pop - (analysis of generational issue changes in k-pop idol groups - focusing on news big data). *Korean Journal of Popular Culture*.

Grootendorst, M. (2022). Bertopic: Neural topic modeling with a class-based tf-idf procedure. *arXiv preprint arXiv:2203.05794*.

Kim, J. et al. (2017). The attribute agenda-setting influence of online community on online newscast: Investigating the south korean sewol ferry tragedy. *Asian Journal of Communication*.

Lee, S. and Park, J. (2023). The collapse and rebuilding of k-pop idols' persona. *Journal of Cultural Studies*.

McCombs, M. (2006). Agenda-setting effects and the media. *Journalism & Mass Communication Quarterly*, 83(2):205–215.

Park, C. (2017). Online communities and information diffusion in south korea. *Korean Finance Forum*.

Smith, J. and Doe, A. (2024). Topic granularity & hallucination in llm topic modelling. *arXiv preprint arXiv:2405.00611*.

Waltman, L. and van Eck, N. J. (2018). Granularity of algorithmically constructed publication-level classifications of research publications: Identification of topics. *arXiv preprint arXiv:1801.02466*.

Wang, X. et al. (2021). Multi-granularity learning towards open topic classification. *Information Sciences*, 580:435–448.

Wolf, T., Debut, L., Sanh, V., Chaumond, J., Delangue, C., Moi, A., Cistac, P., Rault, T., Louf, R., Funtowicz, M., et al. (2019). Huggingface's transformers: State-of-the-art natural language processing. *arXiv preprint arXiv:1910.03771*.